

Wave Selection at an $O(2)$ -Hopf Bifurcation in Conservative Two-Component PDE Systems

Part I

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Plan

- ▶ **Part I.** The $O(2)$ -Hopf bifurcation: traveling and standing waves, the classification by two cubic coefficients ζ, ξ , and closed-form formulas for ζ, ξ for a class of conservative two-component PDEs.
- ▶ **Part II.** A structural property of the nonlinearity that forces $\xi_R = 0$ for every admissible linear part.

Guiding question: given a PDE, which wave appears at onset, and is it stable?

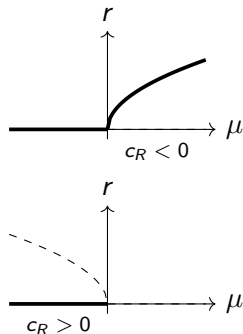
Hopf bifurcation in one complex variable

$$\dot{z} = (\mu + i\omega)z + c|z|^2z, \quad z \in \mathbb{C}, \quad c = c_R + ic_I.$$

In polar coordinates $z = re^{i\theta}$:

$$\dot{r} = \mu r + c_R r^3, \quad \dot{\theta} = \omega + c_I r^2.$$

- ▶ Periodic orbit $r^2 = -\mu/c_R$, existing for $\mu c_R < 0$.
- ▶ $c_R < 0$: supercritical, the orbit is stable.
- ▶ $c_R > 0$: subcritical, the orbit is unstable.
- ▶ c_I only shifts the frequency.



Only the sign of one real number, c_R , decides the outcome.

What $O(2)$ symmetry changes

On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, the group $O(2)$ acts by translations $x \mapsto x + h$ and the reflection $x \mapsto -x$.

- ▶ A critical mode $e^{im_c x}$ comes with $e^{-im_c x}$; each of $\pm i\omega_c$ becomes a double eigenvalue.
- ▶ The center subspace is four-dimensional, with two complex amplitudes:

$A e^{i(m_c x + \omega_c t)}$ and $B e^{i(m_c x - \omega_c t)}$: waves moving in opposite directions.

- ▶ Symmetry forces the cubic normal form

$$\begin{aligned}\dot{Z}_1 &= \beta(\lambda)Z_1 + Z_1(\zeta|Z_1|^2 + \xi|Z_2|^2), \\ \dot{Z}_2 &= \overline{\beta(\lambda)}Z_2 + Z_2(\bar{\xi}|Z_1|^2 + \bar{\zeta}|Z_2|^2).\end{aligned}$$

Two real numbers now decide the outcome: $\zeta_R = \operatorname{Re} \zeta$ (self-coupling) and $\xi_R = \operatorname{Re} \xi$ (cross-coupling).

The class of systems

$$\frac{d}{dt} \begin{bmatrix} u \\ v \end{bmatrix} = L_\lambda \begin{bmatrix} u \\ v \end{bmatrix} + G \begin{bmatrix} u \\ v \end{bmatrix}, \quad x \in \mathbb{T}, \quad \int_{\mathbb{T}} u = \int_{\mathbb{T}} v = 0.$$

- Reflection: $R(u, v)(x) = (u(-x), -v(-x))$. Equivariance forces

$$L_{11}, L_{22} \text{ even in } \partial_x, \quad L_{12}, L_{21} \text{ odd in } \partial_x.$$

- Nonlinearity: $g_i = \sum a_{\alpha_1, \alpha_2}^i (Du)^{\alpha_1} (Dv)^{\alpha_2}$, with $(Du)^\alpha = \prod_j (\partial_x^j u)^{\alpha_j}$;
example: $\alpha = (2, 0, 1, 0, \dots)$ gives $u^2 u_{xx}$.
- Conservative: $g_i = \partial_x f_i$.
- Fourier symbol M_m ($\partial_x \mapsto im$); eigenvalues $\beta_{m,1}, \beta_{m,2}$ solve

$$\beta^2 - \operatorname{tr} M_m \beta + \det M_m = 0.$$

Hypotheses

Assumption 1 (Linear part)

Principal part of order $2m_L$, diagonal and nondegenerate; λ enters below order $2m_L$.

Assumption 2 (Nonlinearity)

G is an equivariant, conservative differential polynomial, of order $r_G \leq 2m_L - 1$.

Assumption 3 (Hopf crossing at (m_c, λ_c))

- (i) $\operatorname{tr} M_{m_c}(\lambda_c) = 0$;
- (ii) $\det M_m(\lambda_c) > 0$ for all $m \neq 0$;
- (iii) $\frac{d}{d\lambda} \operatorname{tr} M_{m_c}(\lambda_c) \neq 0$;
- (iv) $\operatorname{tr} M_m(\lambda_c) < 0$ for $m \neq 0, \pm m_c$;
- (v) $\liminf_{m \rightarrow \infty} \det M_m(\lambda_c) / |\operatorname{tr} M_m(\lambda_c)| > 0$.

The spectral gap

Theorem 1

Under (i)–(iv), the noncritical spectrum satisfies $\sup \operatorname{Re} \beta_{m,n}(\lambda_c) < 0$ if and only if (v) holds.

Each 2×2 block: $\operatorname{Re} \beta < 0$ for both roots iff $\operatorname{tr} M_m < 0 < \det M_m$. The slower root behaves like $\det M_m / \operatorname{tr} M_m$ when this ratio is small.

Condition (v) cannot be dropped. $L_{11} = -\partial_x^4 + \lambda$, $L_{22} = 0$, $L_{12} = L_{21} = \partial_x$:

$$\operatorname{tr} M_m = -m^4 + \lambda, \quad \det M_m = m^2, \quad m_c = 1, \quad \lambda_c = 1,$$

$$\frac{\det M_m}{|\operatorname{tr} M_m|} = \frac{m^2}{m^4 - 1} \rightarrow 0, \quad \beta_{m,2} \sim -m^{-2} \rightarrow 0^-.$$

(i)–(iv) hold, but the eigenvalues accumulate at 0.

Center manifold and normal form

Theorem 2

Under the three assumptions, for λ near λ_c there is a four-dimensional, $O(2)$ -equivariant, locally attracting center manifold, on which a near-identity change of coordinates gives

$$\dot{Z}_1 = \beta_{m_c,1}(\lambda)Z_1 + Z_1(\zeta|Z_1|^2 + \xi|Z_2|^2) + O(|\lambda - \lambda_c||Z|^3 + |Z|^5),$$

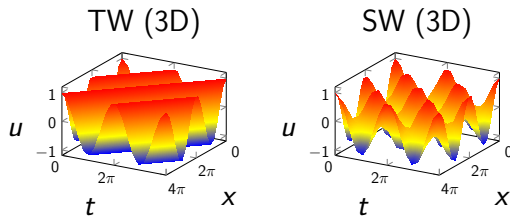
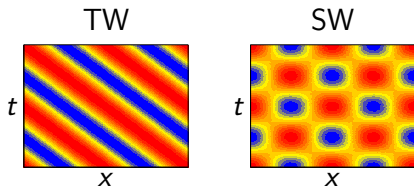
and the equation for Z_2 obtained by symmetry, with

$$\zeta = \hat{s}_{1,1} + \hat{c}_{1,1}, \quad \xi = \hat{s}_{1,2} + \hat{c}_{1,2}.$$

- ▶ $\hat{s}_{1,j}$: *self-interaction*, linear in the cubic coefficients of G .
- ▶ $\hat{c}_{1,j}$: *cross-interaction*, bilinear in the quadratic coefficients of G , via the second harmonic at mode $2m_c$.

Traveling and standing waves

- **TW:** $(Z_1, Z_2) = (re^{i\theta t}, 0)$, $r^2 = -\operatorname{Re} \beta / \zeta_R$; $(u, v)^T \approx 2r \operatorname{Re}(e^{i(m_c x + \theta t)} \mathbf{q}_1)$.
- **SW:** $(Z_1, Z_2) = r(e^{i\theta t}, e^{-i\theta t})$, $r^2 = -\operatorname{Re} \beta / (\zeta_R + \xi_R)$;
 $(u, v)^T \approx 2r \operatorname{Re}(e^{i(m_c x + \theta t)} \mathbf{q}_1 + e^{i(m_c x - \theta t)} \mathbf{q}_2)$.



A TW breaks the reflection symmetry and translates; a SW keeps it and has fixed nodes.

Radial dynamics

With $r_j = |Z_j|$ and $\mu = \operatorname{Re} \beta(\lambda)$:

$$\dot{r}_1 = r_1(\mu + \zeta_R r_1^2 + \xi_R r_2^2), \quad \dot{r}_2 = r_2(\mu + \xi_R r_1^2 + \zeta_R r_2^2).$$

TW $(r_1, r_2) = (r, 0)$:

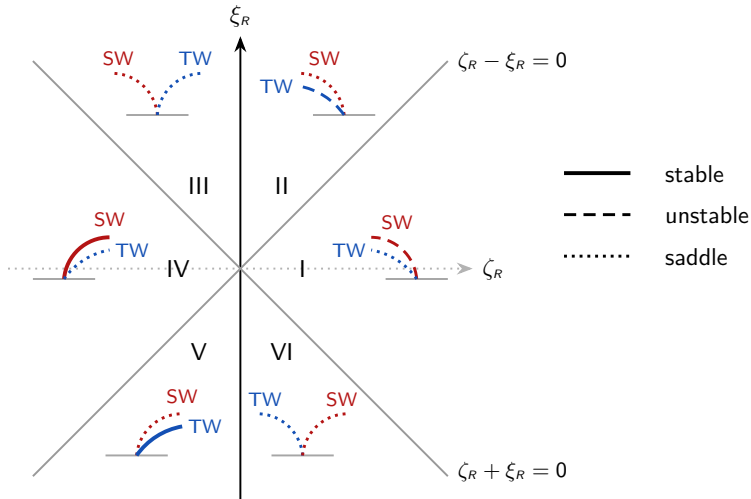
eigenvalues $-2\mu, \quad \mu\left(1 - \frac{\xi_R}{\zeta_R}\right).$

SW $r_1 = r_2 = r$:

eigenvalues $2r^2(\zeta_R + \xi_R), \quad 2r^2(\zeta_R - \xi_R).$

- ▶ TW is stable iff $\xi_R < \zeta_R < 0$; SW is stable iff $\zeta_R < -|\xi_R|$.
- ▶ The lines $\zeta_R = 0$, $\zeta_R + \xi_R = 0$, $\zeta_R - \xi_R = 0$ divide the (ζ_R, ξ_R) -plane into six regions.
- ▶ Away from these lines the branches and their stability persist under the higher-order terms.

Classification in the (ζ_R, ξ_R) -plane



Branches drawn to the right (left) of each origin correspond to $\text{Re } \beta > 0$ (< 0).

Regions

Region	Conditions	$\text{Re } \beta$	TW	SW
I	$\zeta_R + \xi_R > 0, \zeta_R - \xi_R > 0$	—	saddle	unstable
II	$\zeta_R > 0, \zeta_R - \xi_R < 0$	—	unstable	saddle
III	$\zeta_R < 0, \zeta_R + \xi_R > 0$	—	dne	saddle
		+	saddle	dne
IV	$\zeta_R + \xi_R < 0, \zeta_R - \xi_R < 0$	+	saddle	stable
V	$\zeta_R < 0, \zeta_R - \xi_R > 0$	+	stable	saddle
VI	$\zeta_R > 0, \zeta_R + \xi_R < 0$	+	dne	saddle
		—	saddle	dne

dne: does not exist. Only Regions IV and V have a stable small-amplitude branch (continuous transition); elsewhere the transition is a jump.

Application 1: a regularized p -system

$$\partial_t u - \partial_x v = -a \partial_x^4 u, \quad \partial_t v - \partial_x \sigma(u) = -\delta \partial_x^2 v - \partial_x^4 v, \quad \sigma(u) = c^2 u + \frac{1}{2} u^2 + \frac{\eta}{3} u^3.$$

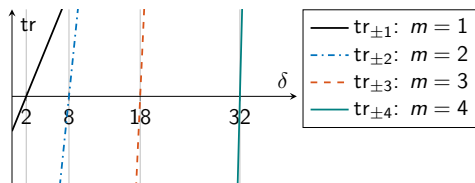
u : deformation gradient, v : velocity; bifurcation parameter δ .

$$M_m = \begin{pmatrix} -am^4 & im \\ ic^2 m & \delta m^2 - m^4 \end{pmatrix},$$

$$\text{tr } M_m = m^2(\delta - (a+1)m^2).$$

Hopf at $m_c = 1$, $\delta_c = a + 1$, $\omega = \sqrt{c^2 - a^2}$,
provided

$$0 < a < c, \quad c^2 > \frac{4a(a+1)^3}{27}.$$



Application 1: coefficients

$$\zeta = \hat{c}_{1,1}(a, c) + \frac{i\eta}{2c^2\omega}, \quad \xi = \frac{i}{c^2\omega} \left[\eta - \frac{1}{2(c^2 - 16a(a - 3))} \right],$$

$$\zeta_R = -\frac{a + 1}{6c^2[a^2(5a - 16)^2 + 4(a + 1)^2(c^2 - a^2)]} < 0, \quad \xi_R \equiv 0.$$

- ▶ Region IV for all admissible (a, c, η) : traveling waves are saddles, standing waves are stable.
- ▶ The cubic stress coefficient η enters only the imaginary parts.
- ▶ As $a \rightarrow 0^+$ the spectral gap closes, but the coefficients extend continuously: $\zeta_R \rightarrow -1/(24c^4)$.

Question 1

Is $\xi_R \equiv 0$ a consequence of conservativity?

Application 2: a conservative bilinear family

Same linear part, the most general conservative equivariant bilinear nonlinearity of first order:

$$G(u, v) = \begin{bmatrix} g_{11}(u_x v + u v_x) \\ g_{21} u u_x + g_{22} v v_x \end{bmatrix}.$$

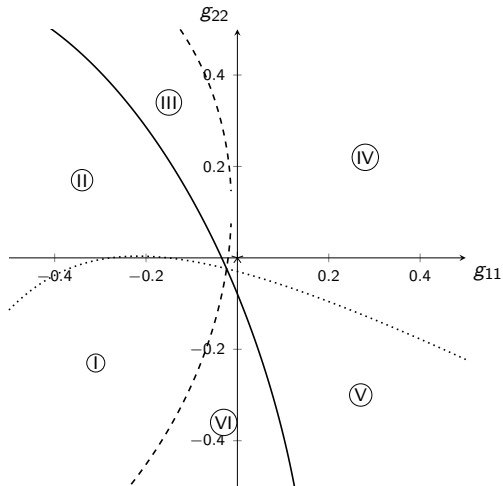
The p -system corresponds to $(g_{11}, g_{21}, g_{22}) = (0, 1, 0)$. Here $\hat{s}_{1,j} = 0$ and

$$\zeta_R = \frac{\mathbf{g}^T N \mathbf{g}}{12c^2 D_\zeta}, \quad \xi_R = -\frac{a(g_{11} - g_{22})[c^2(g_{11} - 4g_{22}) + 4g_{21}]}{c^2 D_\xi},$$

with $D_\zeta = a^2(5a - 16)^2 + 4(a + 1)^2(c^2 - a^2)$, $D_\xi = c^2 - 16a(a - 3)$, and $N = N^T$ explicit.

Conservativity alone does not force $\xi_R = 0$: $\xi_R \neq 0$ as soon as $g_{11} \neq g_{22}$ and the bracket is nonzero.

Application 2: all six regions occur



Slice $a = \frac{1}{2}$, $c = 1$, $g_{21} = 1$.

Solid: $\zeta_R = 0$.

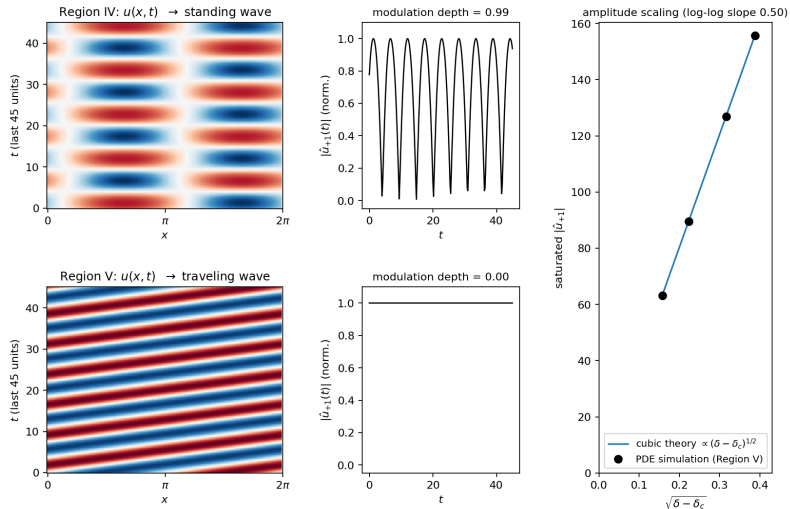
Dashed: $\zeta_R + \xi_R = 0$.

Dotted: $\zeta_R - \xi_R = 0$.

The star at the origin is the p -system, in Region IV.

With $g_{11} = 0$ only Regions IV–VI occur; no two of the three terms realize the full classification.

Direct simulation



Top: Region IV ($g_{11} = g_{22} = 0$), standing wave. Bottom: Region V ($g_{11} = 0.35$, $g_{22} = -0.30$), traveling wave. Right: amplitude $\propto (\delta - \delta_c)^{1/2}$.

Summary of Part I

- ▶ At an $O(2)$ -Hopf bifurcation the selected wave is decided by the signs of ζ_R , $\zeta_R + \xi_R$, $\zeta_R - \xi_R$.
- ▶ For conservative two-component systems, ζ and ξ are given in closed form by the coefficient array and the spectral data of M_{m_c} , M_{2m_c} .
- ▶ The spectral gap is characterized by a condition on the symbol.
- ▶ In the p -system, $\xi_R \equiv 0$ and η is invisible to selection; in the larger conservative bilinear family, every region occurs.

Question 2

Which property of the p -system nonlinearity forces $\xi_R \equiv 0$? (Part II)